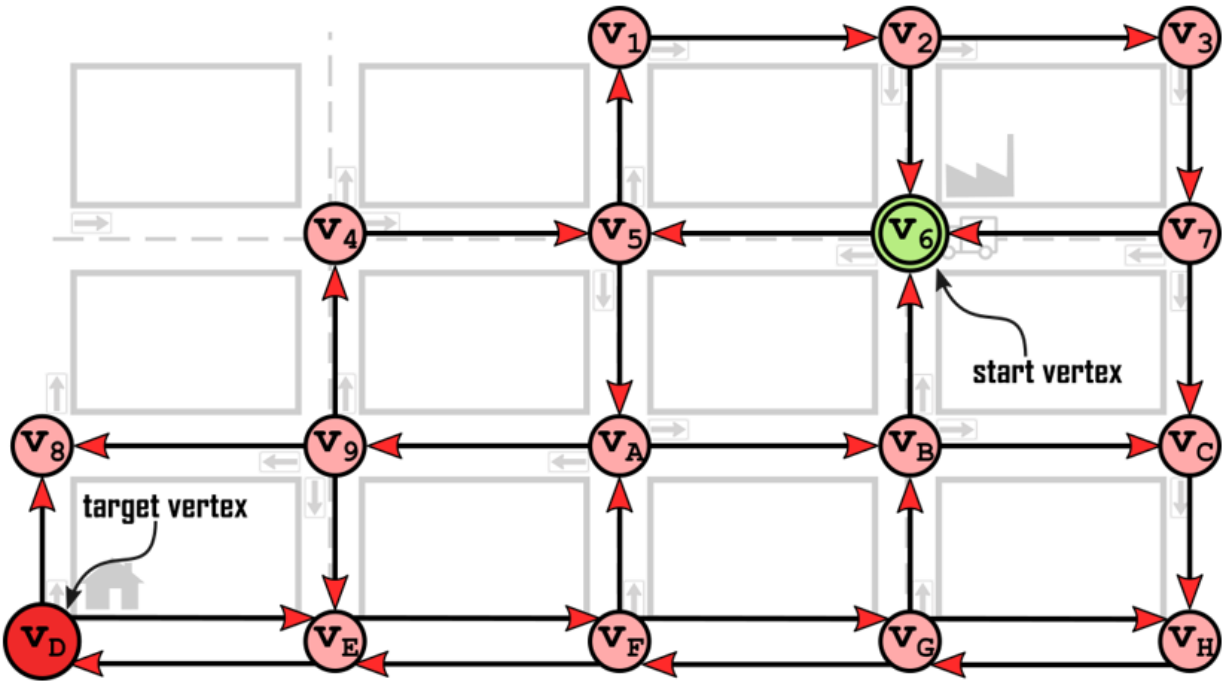



**Figure 14.9 Building a graph on the map in figure 14.8. We added a vertex to each road intersection (some vertices are however omitted for better readability) and edges connect adjacent intersections connected by a one-way road (two-way roads are modeled with a couple of edges).**

Figure 14.9 shows the graph that can be built over the map shown in figure 14.8, where we added a vertex at each road intersection, and an edge going from vertex v to vertex u means that from the intersection modeled with v to the one modeled with u, there is a road that can be traveled in that direction (and not necessarily in the opposite one, from u to v).

If all roads could be traveled in both directions, we would just go West on Market from the warehouse until we crossed $10^{th}$ St., and then South on $10^{th}$ down to our destination.